MSc Data Science

DS601: Mathematical Foundations for Data Science

Tutorial 1 - Complete Worked Solutions (Questions 1-7)

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Question 1: Birthday Problem

A class contains 60 students. Assume 365 equally likely birthdays, independent birthdays, and no leap day. Find the probability that at least two students share a birthday.

1. Concept Used: Complement Rule and Multiplication Rule

Directly counting all ways in which at least one birthday match occurs is difficult because many different types of matches are possible. The complement rule replaces the event by its opposite:

P(at least one shared birthday) = 1 - P(all birthdays are different).

The multiplication rule is then used because the second student must avoid the first birthday, the third must avoid the first two birthdays, and so on.

2. Approach

Calculate the probability that all 60 birthdays are distinct, then subtract that probability from 1.

3. Step-by-Step Solution

Step 1: The first student may have any birthday, so the probability is 365/365 = 1.

Step 2: The second student must avoid the first birthday, giving 364/365.

Step 3: The third student must avoid two occupied birthdays, giving 363/365.

Step 4: Continue in the same manner. For the 60th student, 59 birthdays are already occupied, so the factor is 306/365.

P(all different) = (365/365)(364/365)(363/365)...(306/365).

P(all different) ≈ 0.005877.

P(at least one shared birthday) = 1 - 0.005877 = 0.994123.

Final answer: approximately 99.41%.

Question 2: Assigning President, Vice-President and Secretary

There are 30 students. Three different students are selected and assigned the roles of President, Vice-President and Secretary.

1. Concept Used: Permutation and Multiplication Principle

A permutation is used when order or assigned roles matter. Here, A as President and B as Vice-President is different from B as President and A as Vice-President. Repetition is not allowed because one student cannot occupy two roles.

$$nPr = n!/(n-r)!$$

2. Approach

Count the choices for each role in sequence. The number of available students decreases after each assignment.

3. Step-by-Step Solution

Step 1: President - 30 choices.

Step 2: Vice-President - 29 remaining choices.

Step 3: Secretary - 28 remaining choices.

Total assignments = 30 x 29 x 28 = 24,360.

For the exact assignment A = President, B = Vice-President and C = Secretary, there is one favourable outcome among 24,360 equally likely assignments.

$$P = 1/24,360.$$

Final answer: order matters; repetition is not allowed; total assignments = 24,360; required probability = 1/24,360.

Question 3: Throwing a Four-Sided Die Twice

A fair four-sided die numbered 1, 2, 3 and 4 is thrown twice. Let X be the first result and Y the second result.

1. Concept Used: Sample Space, Ordered Pairs and Classical Probability

Each outcome is an ordered pair (X,Y). Since the two throws are independent and the die is fair, all 4 x 4 = 16 ordered outcomes are equally likely.

P(E) = number of favourable outcomes / total number of equally likely outcomes.

2. Approach

List the 16 ordered pairs, then count those satisfying each event.

3. Step-by-Step Solution

	Y=1	**Y=2**	**Y=3**	**Y=4**
X=1	(1,1)	(1,2)	(1,3)	(1,4)
X=2	(2,1)	(2,2)	(2,3)	(2,4)
X=3	(3,1)	(3,2)	(3,3)	(3,4)
X=4	(4,1)	(4,2)	(4,3)	(4,4)

Part (a): The sample space contains the 16 ordered pairs shown above.

Part (b): X+Y is odd exactly when one result is odd and the other is even. The favourable outcomes are (1,2), (1,4), (3,2), (3,4), (2,1), (2,3), (4,1), (4,3).

P(X+Y is odd) = 8/16 = 1/2.

Part (c): Obtaining 1 and 4 in any order gives (1,4) and (4,1).

P(1 and 4 in any order) = 2/16 = 1/8.

Final answers: sample-space size = 16; P(X+Y odd) = 1/2; P(1 and 4 in any order) = 1/8.

Question 4: Committee Selection from Four Categories

An 8-member committee is selected from 18 candidates in four mutually exclusive categories of sizes 5, 5, 4 and 4. Count committees containing at least one member from every category.

1. Concept Used: Combinations, Complement Principle and Inclusion-Exclusion

A committee is an unordered selection, so combinations are used. Let U be the set of all 8-member committees. Define E_A, E_B, E_C and E_D as the sets of committees missing categories A, B, C and D, respectively. The valid committees are:

$$U \setminus (E_A \cup E_B \cup E_C \cup E_D).$$

The inclusion-exclusion principle is needed because a committee missing two categories lies in two bad sets and would otherwise be subtracted twice.

2. Approach

Count all committees, subtract committees missing one category, and add back committees missing two categories. Triple omissions are impossible because one remaining category has fewer than 8 candidates.

3. Step-by-Step Solution

Step 1: Total unrestricted committees:

$$C(18,8) = 43{,}758.$$

Step 2: Committees missing one category:

Two categories have size 5, so omitting either leaves 13 candidates: $2C(13,8)$. Two categories have size 4, so omitting either leaves 14 candidates: $2C(14,8)$.

$$2C(13,8) + 2C(14,8) = 8{,}580.$$

Step 3: Add back pairwise intersections:

Omit both 5-member categories: $C(8,8)=1$. Omit one 5-member and one 4-member category: there are 4 such pairs, each giving $C(9,8)=9$, so 36. Omit both 4-member categories: $C(10,8)=45$.

$$\text{Total pairwise intersections} = 1 + 36 + 45 = 82.$$

Step 4: Triple and quadruple intersections are zero because fewer than 8 candidates remain.

$$\text{Valid committees} = 43{,}758 - 8{,}580 + 82 = 35{,}260.$$

Final answer: 35,260 committees.

Question 5: Participation in Cricket, Football and Chess Clubs

Among 200 students: $|C|=120$, $|F|=90$, $|H|=80$, $|C\cap F|=50$, $|C\cap H|=40$, $|F\cap H|=30$ and $|C\cap F\cap H|=20$.

1. Concept Used: Inclusion-Exclusion and Disjoint Venn Regions

For three sets, inclusion-exclusion gives:

$|C\cup F\cup H| = |C|+|F|+|H| - |C\cap F| - |C\cap H| - |F\cap H| + |C\cap F\cap H|$.

The pairwise intersection counts include students in all three clubs. Therefore, to obtain students in exactly two clubs, subtract the triple intersection from each pairwise intersection.

2. Approach

First find the union. Then calculate exactly-two regions, exactly-one regions, and finally the complement outside the union.

3. Step-by-Step Solution

Part (a): At least one club

$|C\cup F\cup H| = 120+90+80-50-40-30+20 = 190$.

Part (b): Exactly one club

Cricket only = $120-50-40+20 = 50$.

Football only = $90-50-30+20 = 30$.

Chess only = $80-40-30+20 = 30$.

Exactly one = $50+30+30 = 110$.

Part (c): Exactly two clubs

Cricket and Football only = $50-20 = 30$.

Cricket and Chess only = $40-20 = 20$.

Football and Chess only = $30-20 = 10$.

Exactly two = $30+20+10 = 60$.

Part (d): No club

$|U \setminus (C\cup F\cup H)| = 200-190 = 10$.

Final answers: at least one = 190; exactly one = 110; exactly two = 60; no club = 10.

Question 6: License Plates with at Least One Vowel and One Even Digit

A license plate has 3 letters followed by 3 digits. Repetition is allowed. Count plates containing at least one vowel among the letters and at least one even digit among the digits.

1. Concept Used: Complement Principle and Multiplication Principle

For an “at least one” condition, it is easier to count all possibilities and subtract those with none. The letter block and digit block are independent, so their valid counts are multiplied.

2. Approach

Count 3-letter strings containing at least one vowel. Separately count 3-digit strings containing at least one even digit. Multiply the two results.

3. Step-by-Step Solution

Letters: There are 26 letters, of which 5 are vowels and 21 are consonants.

All 3-letter strings = 26^3 = 17,576.

Strings with no vowel = 21^3 = 9,261.

Strings with at least one vowel = 17,576-9,261 = 8,315.

Digits: There are 10 digits. The even digits are 0,2,4,6,8, so the five non-even choices are 1,3,5,7,9.

All 3-digit strings = 10^3 = 1,000.

Strings with no even digit = 5^3 = 125.

Strings with at least one even digit = 1,000-125 = 875.

Required plates = 8,315 x 875 = 7,275,625.

Final answer: 7,275,625 license plates.

Question 7: Six-Digit Numbers with Distinct Digits

Count six-digit numbers whose digits are distinct, whose first digit is odd, and which are divisible by 5.

1. Concept Used: Case Analysis and Permutations without Repetition

A number is divisible by 5 exactly when its final digit is 0 or 5. These two possibilities form disjoint cases. Within each case, distinct digits are arranged without repetition.

2. Approach

Split into Case 1: last digit 0, and Case 2: last digit 5. Count each case carefully, then add because the cases cannot occur simultaneously.

3. Step-by-Step Solution

Case 1: Last digit is 0.

The first digit must be odd: 1,3,5,7 or 9, giving 5 choices.

After fixing the first and last digits, 8 digits remain. Fill the four middle positions with distinct digits:

8P4 = 8 x 7 x 6 x 5 = 1,680.

Case 1 total = 5 x 1,680 = 8,400.

Case 2: Last digit is 5.

The first digit must be odd but cannot be 5, so it has 4 choices: 1,3,7 or 9.

Again, after fixing first and last digits, 8 digits remain for the four middle positions:

8P4 = 1,680.

Case 2 total = 4 x 1,680 = 6,720.

Total = 8,400 + 6,720 = 15,120.

Final answer: 15,120 six-digit numbers.